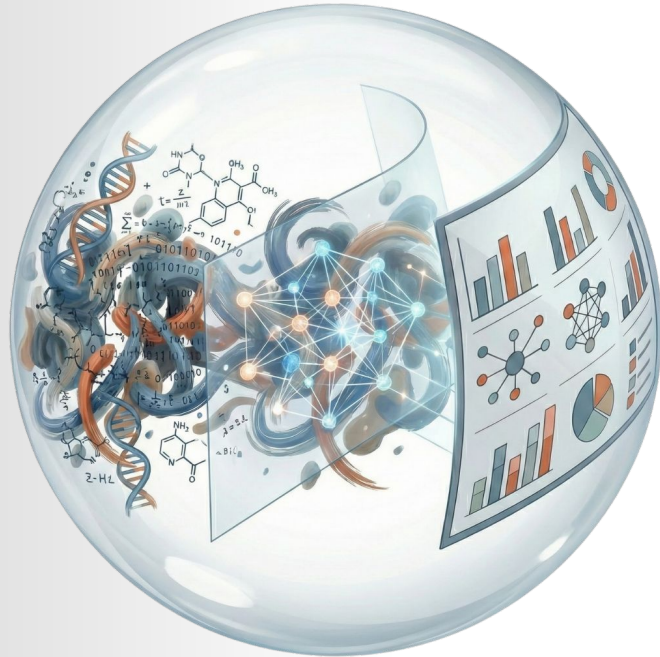




Introduction to Web Development and *Promptotyping* with Claude Code

Project: Legal History Hub
Max-Planck-Institut für
Rechtsgeschichte & Rechtstheorie
Workshop 2 - 13.03.2026

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Slides were generated AI-assisted.
Images are partly AI-generated.



Workshop Overview

1. Quick Start into Web Development
2. Git
3. AI / Vibe / Agentic ... Coding
4. Promptotyping
5. Live Demo
6. Claude Code 101

Coding with AI - a reasonable Setup

- [VS Code](#) with [Live Server Extension](#)
- Coding Agent:
 - <https://claude.com/product/claude-code>
- Install [Python](#) and <https://nodejs.org/>
- Basic handling of [Browser as Web Development Tool](#)
- [GitHub Account](#) and [GitHub CLI](#)
- Knowledge Organization with .md: <https://obsidian.md/> (optional)
- File transformation and preparation: <https://www.docling.ai/> (optional, “PDF → Markdown”!)

Quickstart: How does the Web work?

Your browser (the client) sends a request to a server. The server sends a response back. The protocol for this is called HTTP.

What comes back are text files: HTML for structure, CSS for appearance, JavaScript for interactivity. Your browser assembles these and displays them as a web page.

URL: The address of a resource on the web. <https://example.com/page.html>

HTTP: Hyper Text Transfer Protocol. The protocol browsers and servers use to communicate. Request → Response.

Client: Your browser. It sends requests and displays the results.

Server: A computer somewhere on the internet that serves files. It waits for requests and responds.

HTML, CSS, JS: The three languages your browser understands. Exactly the ones you're learning this semester, and exactly the ones your coding agents will write for you.

Internet vs. Web

1. **Internet:** the infrastructure. Cables, routers, computers, connected worldwide.
2. **Web:** a service that runs on top of the internet. Web pages, links, browsers. What you're building.
3. The internet has existed since the 1960s. The web since 1991.

URL

The address of a resource on the web.

`https://example.com:443/path/page.html?search=cat#result`

1. `https` – the protocol. The `s` stands for "secure": the connection is encrypted. Without it (`http`), everything is transmitted in plain text. Today, `https` is the standard.
2. `example.com` – the domain (resolved to an IP address via DNS)
3. `:443` – the port (usually invisible because it's the default)
4. `/path/page.html` – the path to the file
5. `?search=cat` – a parameter
6. `#result` – an anchor, a jump target on the page

Frontend und Backend

1. **Frontend:** What happens in the browser. HTML, CSS, JavaScript.
This is what your users see.
2. **Backend:** What happens on the server. Databases, logic, authentication. Invisible to the user.
3. **Our project:** Frontend. But Claude Code can also write backend code if another project requires it.

Is HTML a programming language?

1. **No.** HTML describes structure; it doesn't compute anything. It is a markup language, just like Markdown or XML.
2. **Programming languages:** JavaScript, Python, Java ... Markup languages: HTML, XML, Markdown ...
3. **The distinction:** HTML tells the browser *what to display*. JavaScript tells it *what to do*.

What is Git and why do you need it?

- Version control for code. Every change is saved; nothing is lost.
- You can go back to any previous state at any time. You can see what changed and when. Multiple people can work on the same project without overwriting each other's work.
- Git runs locally on your machine. GitHub/GitLab is the platform where your project lives online ("remote").
- Your coding agents (Claude Code, Copilot, Codex) all work with Git. Commits, branches, pull requests: agents often handle these automatically for you, but you need to understand what's happening.

Git: Key Concepts

- **Repository (Repo):** Your project folder, tracked by Git. Contains the entire history.
- **Clone:** Copying a repo from GitHub ("remote") to your local machine.
- **Staging:** Marking changes to be included in the next commit. Like a shopping cart before checkout.
- **Commit:** A saved snapshot. Includes a short description of what changed.
- **Push:** Uploading your commits from your machine to GitHub ("remote").
- **Pull:** Fetching changes from GitHub to your local machine.
- **Branch:** A parallel line of work. You work on a feature without affecting the main codebase.
- **Merge:** Combining a branch back into the main branch.
- **Pull Request (PR):** A request on GitHub to merge your branch. Others can review and comment on the code before it's merged. ("Merge Request" on GitLab)

<https://git-scm.com/install/>



--distributed-even-if-your-workflow-isnt

🔍 Type / to search entire site...



About

Learn

Tools

Reference

Install

Community

Install

Latest version: 2.53.0 ([Release Notes](#))

Windows

macOS

Linux

Build from Source

Choose your operating system above.

The entire [Pro Git book](#) written by Scott Chacon and Ben Straub is available to [read online for free](#). Dead tree versions are available on [Amazon.com](#).

Coding Agents can install things for you!



What does "AI / Vibe / Agentic ... Coding" mean?

Vibe Coding means trying out LLM-generated code without understanding it in detail.

Informed Vibe Coding (?) adds the ability to evaluate results, spot errors, and make targeted improvements.

“Traditional”

- Writing code, learning syntax, ...
- Investing a lot of time, failing a lot – BUT truly understanding it in the end

LLM-assisted (Agentic Coding)

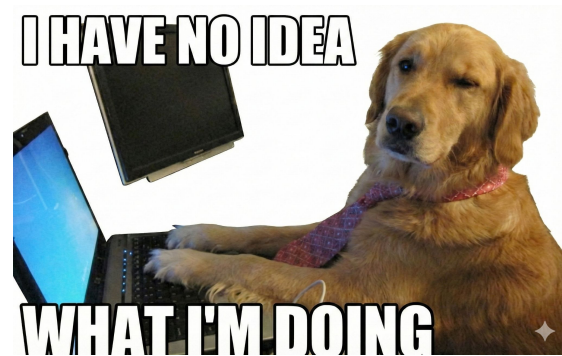
- Formulating requirements, evaluating results, ...
- Very exciting results very quickly – with the lingering feeling: what am I actually doing here?

2025 LLM Year in Review.

<https://karpathy.bearblog.dev/year-in-review-2025/>

Simon Willison's Weblog-

<https://simonwillison.net/2025/Mar/19/vibe-coding>



AI Coding Literacy? – A Proposal

Computational Thinking: Understanding how to solve problems with code

Requirement Engineering: Precisely formulating what you need

Context Engineering: Giving the LLM the right information in its context window

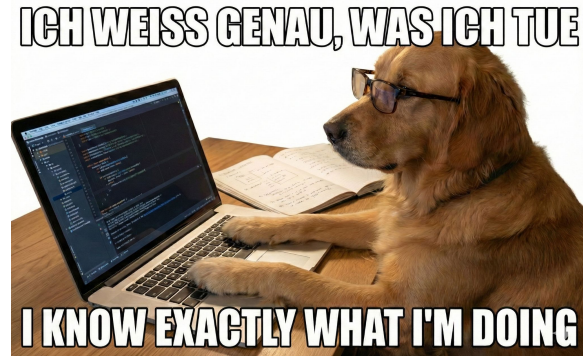
Prompt Engineering: Operationalizing prompting strategies for your own problems

Code Literacy: Being able to read and evaluate generated code

Review: Systematically verifying results

Domain Expertise

...



Computational Thinking

*“Computational thinking involves **solving problems, designing systems, and understanding human behavior**, by drawing on the concepts fundamental to computer science. [...] Thinking like a computer scientist means **more than being able to program a computer**.”*

For working with LLMs, this means: breaking problems down.

Without Computational Thinking: "Build me a website for my data"

With Computational Thinking:

1. Understand the data – What columns are there?
2. Export the data – Excel → CSV
3. Load the data – CSV
4. Display the data – Generate a table
5. Add search – A text field that filters

Wing, J. M. (2006). Computational Thinking.
Communications of the ACM, 49(3), 33-35.
<https://dl.acm.org/doi/10.1145/1118178.1118215>

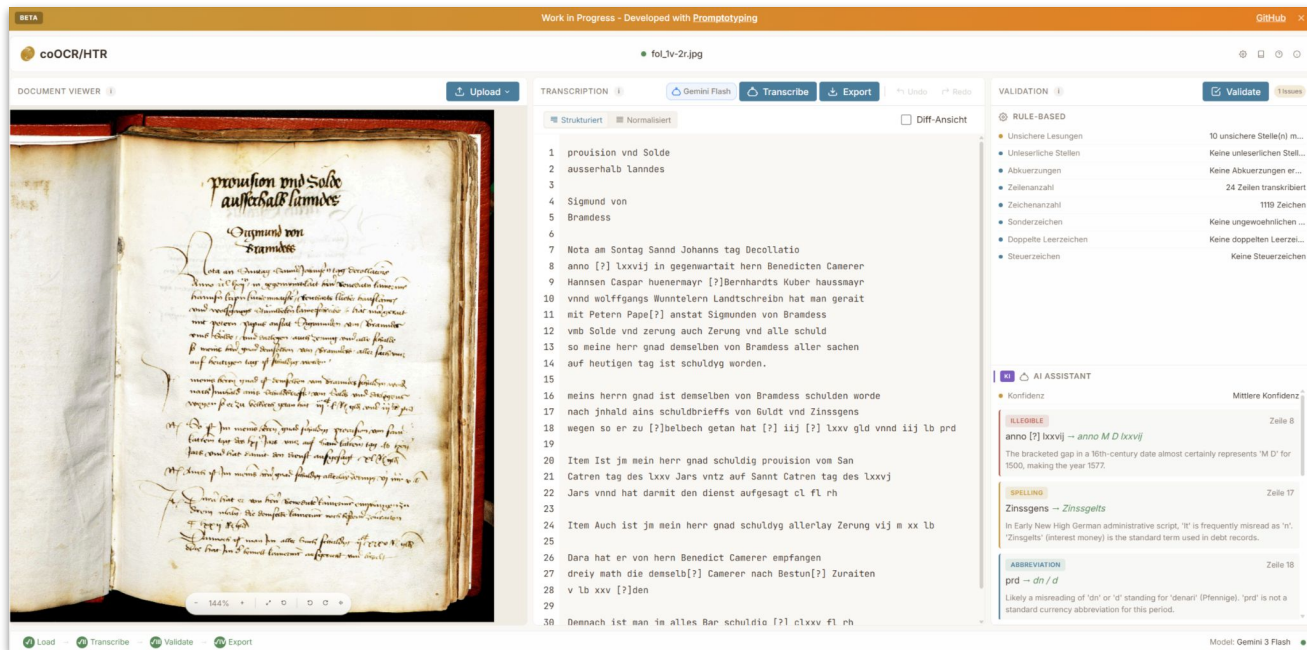
“Roll up your sleeves to not fall behind”

*“There's a **new programmable layer of abstraction** to master involving **agents, sub-agents, prompts, contexts, memory, tools, plugins, skills, hooks ...** and a need to build an all-encompassing **mental model for strengths and pitfalls of fundamentally stochastic, fallible, unintelligible entities.**”*

— Andrej Karpathy, Dezember 2025



coOCR/HTR: Built with *Promptotyping* & Agentic Coding



Editor-in-the-Loop tool for OCR/HTR workflows

Browser-based, open source, open development.

Research Preview (Beta) — built and actively developed using the **Promptotyping** as **Context Engineering** method

Built with **Claude Code + Opus 4.5 & 4.6** Promptotyping

Demo: <https://dhcraft.org/co-ocr-htr>

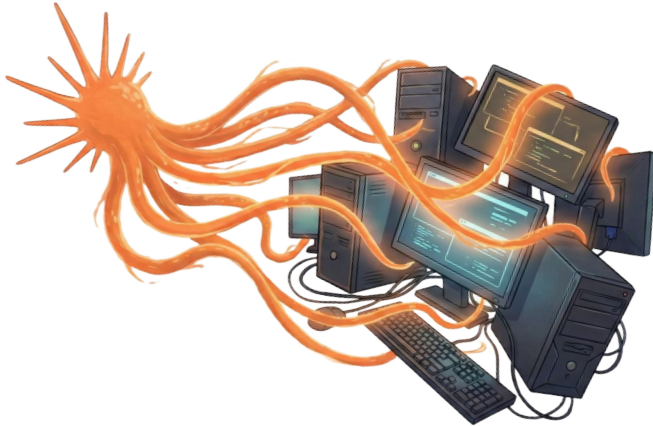
GitHub Repo: <https://github.com/DigitalHumanitiesCraft/co-ocr-htr>

Promptotyping documents: <https://github.com/DigitalHumanitiesCraft/co-ocr-htr/tree/main/knowledge>

Asymmetric Amplification

Why AI Does Not Automate Research —
But Disruptively Amplifies Computer-Based Research
Work.

<https://dhcraft.org/excellence/blog/Asymmetric-Amplification>



Automation Narrative

“If I look at coding, programming, which is one area where AI is making the most progress, what we are finding is we are not far from a world – I think we’ll be there in three to six months – where AI is writing 90% of the code.”

And then in 12 months, we may be in a world where AI is writing essentially all of the code.

— **Dario Amodei**. “The Future of U.S. AI Leadership with CEO of Anthropic Dario Amodei.” <https://www.youtube.com/live/esCSpbDPJik>

Dario Amodei — “We are near the end of the exponential”. <https://youtu.be/N5JDzS9MQYI>

The Two Best AI Models/Enemies Just Got Released Simultaneously.
https://youtu.be/1PxEziv5XIU?si=vmVMA4n_48Cp47sR

Claude AI Co-founder Publishes 4 Big Claims about Near Future: Breakdown. .
<https://youtu.be/lar4vweKGol?si=IBsG9lZWWtzVtOWb>

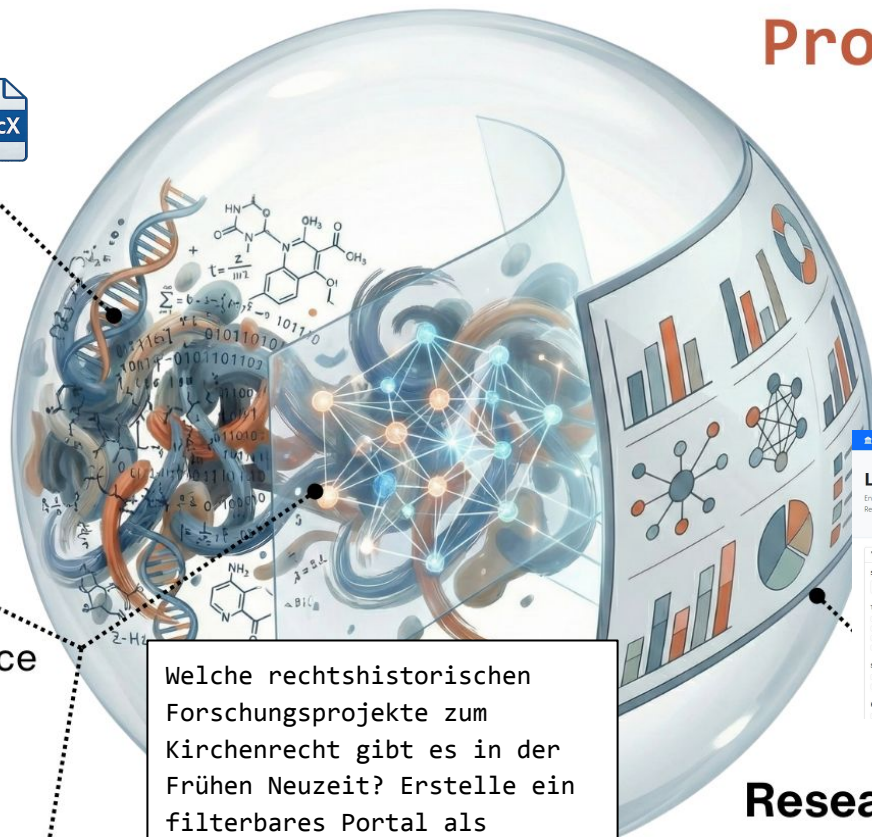
Research Data



Research Domain & Expert in the Loop

Co-Intelligence
(Mollick)

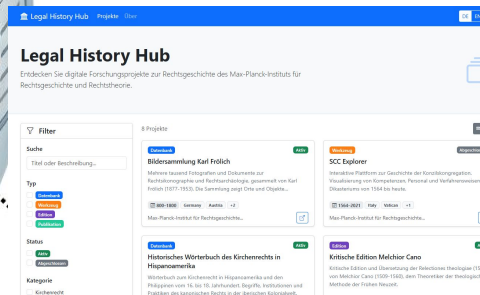
(Frontier-)LLM &
Context Engineering



Welche rechtshistorischen Forschungsprojekte zum Kirchenrecht gibt es in der Frühen Neuzeit? Erstelle ein filterbares Portal als statische Website auf Basis dieser CSV-Metadaten.

Promptotyping

Research Artefacts
(e.g. tools, workflows, models)



PREPARATION



Data & Sources

Context



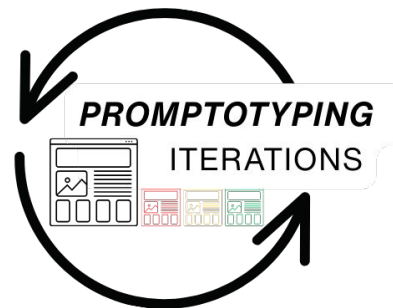
DISTILLATION



DATA.md

REQUIREMENT.md

RESEARCH.md



EXPLORATION & MAPPING

IMPLEMENTATION



Promptotyping - extremely fast, researcher-centred, and research-data-driven creation of prototypes for research tools, workflows, and data-models using frontier LLMs and Context Engineering techniques.

Context Window, Context Rot and Distillation

Context Window – The maximum number of tokens an LLM can process in a single pass. Functions like a working memory that exists only for the duration of the interaction. Claude Opus 4.5 processes up to 200,000 tokens, but the limit is not the goal.

Vaswani, Ashish, et al. "Attention Is All You Need". Advances in Neural Information Processing Systems, Bd. 30, 2017.
<https://arxiv.org/abs/1706.03762>

Context Rot – LLM performance degrades as context length increases. Unstructured text acts as noise, diverting attention from relevant information. More context does not automatically mean better results.

Hong, Kelly, Anton Troynikov, and Jeff Huber. Context Rot: How Increasing Input Tokens Impacts LLM Performance. Chroma, 2025.
<https://research.trychroma.com/context-rot>

Context Engineering – The systematic design of what goes into the context window. Selection, compression, and arrangement of information with the goal of maximizing model response quality under limited resources.

Mei, Lingrui, Jiayu Yao, Yuyao Ge, et al. "A Survey of Context Engineering for Large Language Models". arXiv:2507.13334. Preprint, arXiv, 21. Juli 2025. <https://doi.org/10.48550/arXiv.2507.13334>

Distillation (Context Compression) – The practical application of context engineering. Knowledge is condensed into compact Markdown documents. Maximum information with minimal tokens.

Anthropic. "Effective context engineering for AI agents". Anthropic Engineering Blog, September 2025.
<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

Critical Expert in the Loop

LLMs have no internal mechanisms for verification. Responsibility for correctness lies with the person working with the model. Critical evaluation requires domain knowledge. Only those who know the subject matter can judge whether the model responds plausibly or confabulates.

Pollin, Christopher. 'Promptotyping mit Claude Sonnet 4. Vibe-Coding erfordert den Critical-Expert-in-the-Loop'. Digital Humanities Craft - Research Blogs, 27 May 2025.
<https://dhcraft.org/excellence/blog/Critical-Vibing-Claude-4/>.

Structural weaknesses

- **Sycophancy:** LLMs tend to agree with users rather than contradict them.
- **Confabulation (hallucinations):** LLMs generate plausible-sounding but fabricated details.
- **Non-determinism:** The same prompt yields different results. Variance is inherent to the system, not a bug.

Malmqvist, Lars. „Sycophancy in Large Language Models: Causes and Mitigations“. Preprint, 22. November 2024.
<https://arxiv.org/abs/2411.15287v1>.

Ji, Ziwei, et al. „Survey of Hallucination in Natural Language Generation“. ACM Computing Surveys, Bd. 55, Nr. 12, 2023.
<https://arxiv.org/abs/2202.03629>.

Questions to ask during work

- Did the model challenge my assumption, or merely confirm it?
- Can I independently verify specific claims such as numbers, names, or sources?
- Did I ask in a leading way, or did I leave room for disagreement?
- What happens if I submit the same prompt again?

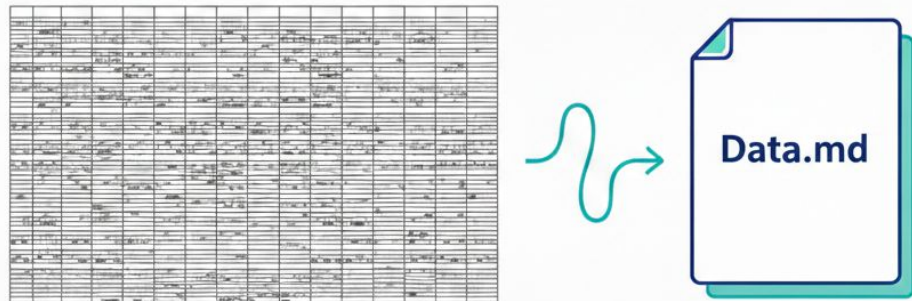
Metacognition – The Critical Expert in the Loop reflects on both output and process.

Context Engineering: Distilling Knowledge into Markdown Documents

Instead of explaining everything to the LLM from scratch in every prompt, we distill our knowledge about a dataset into a structured Markdown document. This document is provided to the LLM as context.

Why?

- Reusable across multiple chats (prevents Context Rot)
- Compact and token-efficient
- Forces us to understand our own data (Expert-in-the-Loop)
- Easy to do and effective!



Context Files for Agentic Systems

[AGENTS.md](#) - An open Markdown format for steering coding agents, maintained under the Linux Foundation. Contains project-specific context: build commands, test requirements, style guidelines. Tool-agnostic, supported by Claude Code, Cursor, Copilot, Gemini CLI, Windsurf, and others.

[CLAUDE.md](#) - Configuration file for Claude Code in the repository root. Defines code style, review criteria, and project rules. Loaded into the system prompt at session start; hierarchical from enterprise to subdirectory level.

[GEMINI.md](#) - Google's equivalent for Gemini Code Assist. Same principle, vendor-specific filename.

All formats follow the same pattern: distill project knowledge into Markdown that structures the agent's context. AGENTS.md is the emerging cross-tool standard; the others are vendor-specific. In practice, teams symlink between them.

Promptotyping-Documents

```
knowledge/  
├── data.md  
├── design.md  
├── implementation.md  
├── journal.md  
├── knowledge.md  
└── requirements.md
```

Promptotyping Documents are Markdown files that distill knowledge about research data, domain, and requirements. They are created during the Exploration and Distillation phases and serve as compact, curated context for working with LLMs.

1. Preparation: Gather all relevant materials: data, documentation about the data, standards, models, research questions, domain knowledge, secondary literature, etc.

2. Exploration & Mapping: Use the LLM to explore what can be done with these data and this context. Make connections between research questions and data structures explicit. The result is knowledge, not documents.

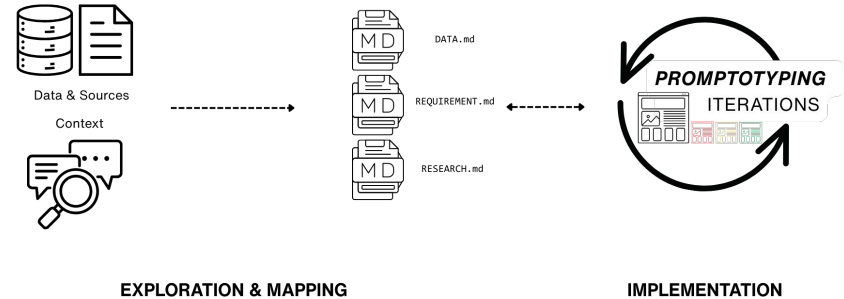
3. Distillation: Condense maximum information with minimal tokens into Promptotyping Documents. The knowledge from Preparation and Exploration is compressed into curated Markdown files.

4. Implementation: Generate a prototype using the Promptotyping Documents as context. The LLM generates, the Critical Expert validates. New knowledge feeds back into the documents.

The 4 Phases of Promptotyping

PREPARATION

DISTILLATION



Element	Markdown Syntax
Heading	# H1 ## H2 ### H3
Bold	**bold text**
Italic	<i>*italicized text*</i>
Blockquote	> blockquote
Ordered List	1. First item 2. Second item 3. Third item
Unordered List	- First item - Second item - Third item
Code	`code`
Horizontal Rule	---
Link	[title](https://www.example.com)
Image	![alt text](image.jpg)

<https://www.markdownguide.org>

Markdown

Lightweight markup language for text formatting, ensuring maximum readability in raw format and easy conversion to structured formats (HTML, PDF, LaTeX ...).

Good for LLMs because:

- High presence in training data
- Token-efficient
- Separates structure from content (e.g., expressing subsections or hierarchies using lists)
- Natively used by Companies (see Agent.md and Skills.md ...)



Live Demo

Promptotyping

Extract information from source and “compress” source structure into a DATA.md

Analyze this excerpt from the Laban Morey Wheaton Day Book (1828-1832), a historical accounting ledger from Norton, Massachusetts.

```
[1]
Monday September-15th-1828
Settled Derius Drake Dr 1 50
To ex divid By Puffer 9/1
Settled By cutting two sticks Cr 189
Puffy Puffer Cr 1 50
By 1 ax div.d. Drake 9/ 342
Asa Danforth Cr
By drawing two logs from Rogers
Monday Sept. 22d 1828
Settled Asa Danforth Cr
By 1 Days work with oven, wheels, &
Settled Derius Drake D 50
To order upon T. Smith for 3/
"
By 1 Days work Cr
October 3d 1828
25 Thomas Danforth 2d
To 1 lb. 4d Cut Nails .8
7 1 Bus. Rye 90t 90 98
Settled Nathaniel Lincoln D 50
To cash 8/ 342
Asa Danforth C
By drawing two sticks from Mss Bowend
4.
139 John Deane D
To cash 6/ 1 00

[...]
```

Extract and structure all entities (like persons, goods/services, places, dates, {what else do exist}), identify all transactions (type, direction, amount, status, {what else do exist}), map the relationships, and note any ambiguities requiring historical interpretation. Use a compact and formal style.

Follow-Up Prompt

How can we accurately, concisely and formally represent the structure of the source in a Markdown document? It needs to represent all information.

Describe the data in a more general way. How can the structure of the source be accurately, concisely and formally represented in a Markdown document? It must represent all the information.

Ist it all correct and perfect? List and explain!

Create the DATA.md.

You have to adjust the follow-up prompts and also control the target. It can be useful to have examples or to describe it in more general terms.

Hands-On: Acquire (a lot of) Data, Explore, Generate [data.md](#) and [requirements.md](#)

Dataset: MoMA Collection Data (Museum of Modern Art, New York). Over 160,000 artworks. CC0 license.

github.com/MuseumofModernArt/collection

Step 1. Set up environment. Create a project folder, open it in VS Code. Claude Code is already available in the sidebar.

Step 2. Acquire data. Ask Claude Code: `download Artworks.csv from github.com/MuseumofModernArt/collection into this folder.`

Step 3. Explore data. Ask Claude Code to explore the dataset for you: `analyze Artworks.csv, show me the columns, data types, value ranges, and a few sample rows.` Claude Code will write and run the script itself. No copy-pasting needed.

Step 4. Iterate. Ask follow-up questions directly: `how many artworks have no date? which nationalities appear most often? show me the distribution of mediums.` Claude Code runs each analysis and returns the results directly in the chat.

Step 5. Generate data.md. Ask Claude Code: `based on your exploration, generate a data.md that documents this dataset – structure, columns, value ranges, data quality issues, and notable patterns.`

Step 6. Review the file, ask for corrections: `the Date column also contains date ranges like "1920-1925", add that to data.md.`

Step 67 Explore the possibility space. Ask Claude Code: `read data.md and propose user stories for a web-based research interface, published as a static site with a single index.html using vanilla JavaScript and CSS. Write the result as requirements.md. Review and refine.`

By the end of this hands-on you have two knowledge documents: a `data.md` that describes your dataset and a `requirements.md` that maps what could be built with it. We will use these to create a web tool later.

```
# DATA.md - MOMA Collection Artworks

## 1. Identität und Herkunft

**Quelle.** Museum of Modern Art (MOMA), New York. Öffentlich zugänglicher Sammlungsdatensatz.
**Model.** Artworks.EAV
**Umfang.** Über 160.000 Einträge (exakte Zahl durch Profiling zu bestätigen).
**Granularität.** Eine Zeile pro Sammlungsobjekt. Werkgruppen und ihre Einzelteile können als separate Zeilen klassifizieren (siehe AccessionNumber).
**Lizenz.** Zu prüfen: MOMA veröffentlicht Sammlungsdaten üblicherweise unter CC0 auf https://www.moma.org.

...

## 2. Feldschema

Jedes Feld ist einmal dokumentiert, einschließlich Typ, Kodierung und struktureller Besonderheiten. "Typ (CSV)" beschreibt, was Pandas beim Einlesen erkennt, nicht die Semantik.

| Gruppe | Feld | Typ (CSV) | Kodierung und Besonderheiten | |
|---|---|---|---|---|
| **Name** | Title | object | Freitext. Kann Ortangaben, Projektnamen und Untertitel enthalten. Teilweise sehr lang. Inkonsistente Leerzeichen beachtet ("New York, New York, Episode", ObjectID 21). |
| | ObjectID | int64 | Fortlaufend. Stellt ein Primärschlüssel. |
| | AccessionNumber | object | Hierarchisches Schema, z.B. "3.1995.1-24" (Gruppe) vs. "3.1995.1" (Einzelblatt). Die Gruppenpräfixe bezeichnen vermutlich einen Erwerbungsvorgang, die genaue Nummerierung liegt oft aber nicht belegt. **Konsequenz:** Ein physisches Werkensemble kommt sowohl als Gruppe als auch in Einzeleinträgen vor. Ob Gruppen- oder Einzeleinträge geparkt werden, muss pro Analyse entschieden werden. |
| | URL | object | Aufbau: https://www.moma.org/collection/works/ObjectID/. Vollständiger Weblink in Sample. |
| | ImageURL | object | Link zum Digitalisat. Nicht für alle Werke vorhanden. |
| | Museum | Artist | object | Name. Übernormalisiert, pro Zeile wiederum. Für personengemeine Analysen empfiehlt sich Normalisierung über ConstituentID. Ob bei Kollaborationen mehrere Namen oder Zeilen vorliegen, ist offen. |
| | ConstituentID | int64 | Eindeutige Personen-ID. Ermöglicht Verknüpfung über Werke hinaus. |
| | ArtistBio | object | Zusammengesetzter String, z.B. "(American, born Estonia. 1905-1974)". Enthält Geburtsland und Migrationshintergrund, die in den Einzelfeldern verloren gehen. Für Geburtsland-Analysen ergibt sich als Nationality. |
| | Nationality | object | **Klammerskodierung.** z.B. "(Austrian)". Leere Klammern (") statt NULL bei fehlender Angabe. Klammern müssen beim Parsing entfernt werden. Ob das Vorkäufchen kontrolliert ist, durch Profiling zu klären. |
| | Biography | object | **Klammerskodierung.** Geburtsjahr als String, z.B. "(1861)". "pd.to_numeric()" scheitert an den Klammern; Stringparsing nötig vor Konvertierung. |
| | EndDate | object | **Klammerskodierung + Sentinel.** Todesjahr, z.B. "(1958)". Wert (0) für lebende Künstler*innen statt NULL. Muss als fehlend behandelt werden. |
| | Gender | object | **Klammerskodierung.** z.B. "(male)". In Sample ausschließlich "(male)". Wertesystem von Profiling zu prüfen. |
| | "Beschreibung" | Date | object | **Freitext mit impliziter Struktur.** Beachachtet: Einzeleintrag "(designed 1925, executed 1926)". Für zeitliche Analysen ist ein Parser erforderlich. |
| | Media | object | **Freitext mit impliziter Struktur.** Kommagetrennte Materialien, Träger nach "on". z.B. "Graphite, pen, color pencil, ink, and gouache on tracing paper". Nicht normalisiert. |
| | Dimensions | object | **Freitext, Missformatierung.** Zoll und cm kombiniert. cm-Werte redundant zu Height/Depth. Bei Werkgruppen Präfixe wie "Each: ". |
| | Classification | object | z.B. "Architektur". In Sample komplex, Validierungskontrolle über Gesamtanzahl durch Profiling zu prüfen. |
| | "Verwaltung" | Department | object | z.B. "Architecture & Design". Selbse Einschränkung via Classification. |
| | Creditline | object | Freitext. Angaben, Anlässe, Mischformate. |
| | DateAcquired | object | **Einziges standardisiertes Dateifeld.** Format "YYYY-MM-DD". Sauer nach datetime konvertierbar. |
| | Cataloged | object | Flag. In Sample ausschließlich "Y". Wert "N" plausibel, aber nicht beobachtet. |
| | Online | object | In Sample durchgehend leer. Möglicherweise nicht gefüllt. |
| | "Mater" | Height (cm) | float64 | Befüllt für 2D-Werke in Sample. Ob insgesamt bestesetzt, ist plausibel, aber Sample-Bias (nur Architektur). |
| | Width (cm) | float64 | Same Height. |
| | Depth (cm) | float64 | Dreidimensionale Objekte. In Sample leer. |
| | Circumference (cm) | float64 | In Sample leer. |
| | Diameter (cm) | float64 | In Sample leer. |
| | Length (cm) | float64 | In Sample leer. |
| | Weight (kg) | float64 | In Sample leer. |
| | Seat Height (cm) | float64 | Möbelobjekte. In Sample leer. |
| | Duration (sec.) | float64 | Zeitbasierte Medien. In Sample leer. |

...

**Erwartung zu Mafeldern.** Fillrate stark Classification-abhängig. Mehrzahl der Spalten vermutlich dünn besetzt.

**Fehlende Relationen.** Keine Felder für Ausstellungshistorie, Provenienz (über Creditline hinaus), Werkbeziehungen oder kuratorische Einordnung.

...

## 3. Datenqualitätsprobleme

| Problem | Feld(er) | Beispiel | Status |
| :----- | :----- | :----- | :----- |
| Sentinel statt NULL | EndDate | (0) für Lebende | bestätigt |
| Leere Klammern statt NULL | Nationality | ( ) | bestätigt |
| Klammern verleihten numerisches Parsing | Biography, EndDate, Nationality, Gender | "(1861)" → NaN bei "pd.to_numeric()" | bestätigt |
| Inkonsistente Leerzeichen | Title | "New York, New York, Episode" | bestätigt |
| Werkgruppen-Einzelwerk-Duplikat | AccessionNumber | "3.1995.1-24" neben "3.1995.1" bis ".24" | bestätigt |
| Redundante Feldangaben | Dimensions vs. Height/Width | cm-Werte doppelt | bestätigt |
| Mehrere Künstler*innen pro Werk | Artist, ConstituentID | Kollaborationen | erwartet |
| Fehlende Digitalisate | ImageURL | Lücken | erwartet |
| Logische Maf-Fillrate nach Werkklasse | alle Maf-Spalten | Duration nur Medienkunst | erwartet |
| Unschärfliche Gender-Kodierung | Gender | omnibinary, Gruppen, fehlend | erwartet |
| Schreibvarianten in kategorialen Feldern | Classification, Department | Unkonsequenzen | erwartet |
| Heterogene Date-drumalists | Date | Einzeleintrag vs. Datum je nach Department | erwartet |

...

## 4. Analytisches Potenzial und offene Fragen

**Sammlungspolitik über Zeit.** DataAcquired + Department + Classification. **Offen:** Verteilung von DataAcquired über die Jahrzehnte?

**Demografie der Künstler*innen.** Gender, Nationality, Lebensdaten. **Offen:** Wie viele eindeutige ConstituentIDs? Welche Gender-Werte jenseits von "(male)"?

**Abteilungen und Klassifikationen.** Kreuztabelle Department + Classification. **Offen:** Vollständige Verteilungen und Verteilungen?

**Medien und Materialien.** Längsschnitt nach Parsing von Date und Media. **Einschränkung:** MLP oder regelbasiertes Parsing nötig.

**Zeitliche Verteilung der Werksdatierung.** Erfordert Date-Parser. **Einschränkung:** Circa-Angaben und Spannen erzeugen Unsicherheit.

**Geburtsland vs. Nationalität.** ArtistBio ist reichhaltiger als Nationality. **Einschränkung:** String-Parsing nötig.

**Physische Charakteristika.** Maf-Felder nach Classification. **Offen:** Füllraten der Maf-Spalten pro Werkklasse?

**Werkgruppen-Struktur.** AccessionNumber-Parsing. **Offen:** Wie viele Gruppenbeiträge, und wie verzerrten sie (Zählungen)?

**Digitalisierungsgrad.** ImageURL-Füllrate. **Offen:** Anteil ohne ImageURL? Korrelation mit Department oder Alter?

**Nicht nachbar ohne zusätzliche Daten.** Ausstellungsfrequenz, Marktwert, Provenienzforschung, kuratorische Kontextualisierung.
```

data.md

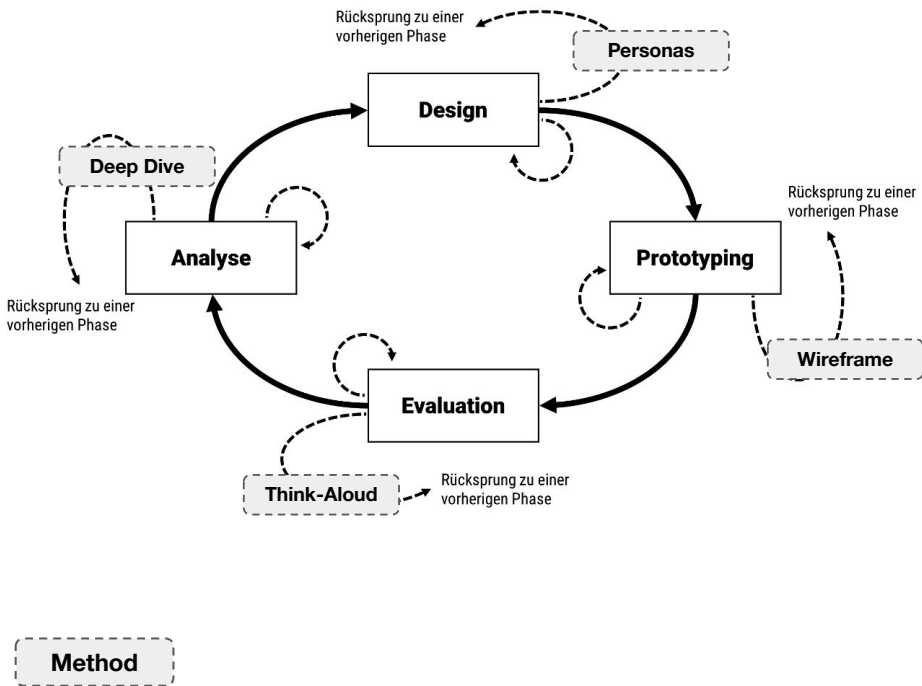
Some Follow-Up Prompts:

- Is everything correct? List and explain.
- Can the same information be presented more compactly without losing detail or complexity? List and explain.
- Write the improved and compact knowledge document data.md

Some Follow-Up Prompts:

- Write the improved and compact knowledge document requirements.md

User-Centred Design → Scholar-Centred Design

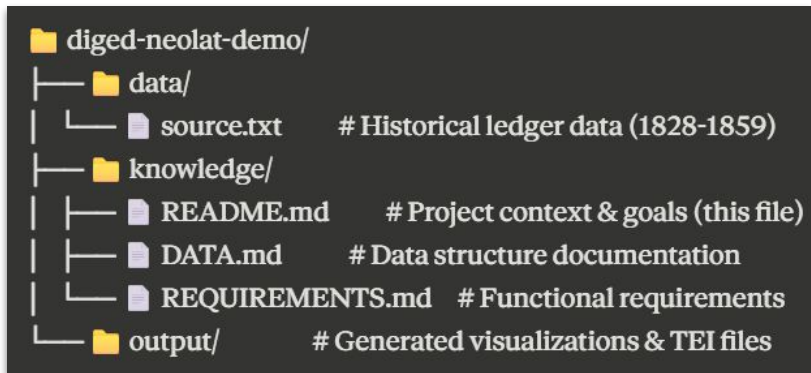


User Stories → Requirements.md

Pollin, Christopher. ‘Modelling, Operationalising and Exploring Historical Information. Using Historical Financial Sources as an Example’. Graz, 2025. <https://resolver.obvsg.at/urn:nbn:at:at-ubg:1-220602>.

As a ...	I want to ...	So that I can ...
social historian	view transaction networks between individuals and organisations in Norton (businesses, churches, schools, and civic institutions)	identify the key economic relationships in the community
social historian	filter networks by personal relationships (gender, marriage, family ties)	find how family connections shaped business patterns
social historian	filter networks by institutional relationships (affiliations, partnerships, roles)	discover how organisational ties influenced economic exchange
social historian	see network changes between 1828-1859	track how community business relationships developed over time
social historian	compare business activities between different community groups	map economic cooperation and division in Norton
social historian	view how men and women participated differently in credit and trade networks	reveal gender patterns in Norton's economic life

Structure



*Promptotyping
Documents*

Domain
Question
Context

Structured Data
Model

Technology Stack
Architecture

Hands-On: Promptotyping in Practice

You created `data.md` and `requirements.md` for the MoMA dataset. In the CorrespExplorer exercise you explored how a Promptotyping Vault translates into a working research tool. Now apply the full Promptotyping methodology yourself to build a simple web-based visualization from scratch.

Step 1 – Set up the Promptotyping Vault. Ask Claude Code: `create a Knowledge subfolder and a Data subfolder in this project. Move data.md and requirements.md into Knowledge. Move artworks.csv into Data.` If you already have design ideas for the interface (layout, color scheme, typography, visual references), create a `design.md` in the Knowledge folder now. If not, you can create one after seeing the first prototype.

Step 2 – Start fresh and provide context. Type `/clear` in Claude Code to reset the conversation. Then ask Claude Code to read your knowledge files before generating anything: `read Knowledge/data.md, Knowledge/requirements.md, and Knowledge/design.md if it exists. Also look at a sample of Data/artworks.csv. Understand the data structure, the requirements, and the goal before writing any code.` This is Context Engineering, not Vibe Coding.

Hands-On: Promptotyping in Practice

Step 3 – Generate a first prototype. Ask Claude Code to **build a static website (single index.html) that loads artworks.csv and visualizes the data.** Claude Code will write the file and you can open it directly in the browser. Do not expect a perfect result. The first iteration is a baseline.

Step 4 – Review as Critical Expert in the Loop. Open **index.html** (Live Server) in your browser. Inspect the output. Apply the patterns from the workshop. Does the HTML reflect the requirements? Are the data quality issues from **data.md** handled correctly? Feed specific errors back into Claude Code. Ask "What is wrong with this?" rather than "Is this good?"

Hands-On: Promptotyping in Practice

Step 5 – Design feedback. Take a screenshot of the generated interface. What works visually, what does not? Create or update your `design.md` with concrete observations: layout changes ("the filter bar should be on the left, not on top"), color and typography preferences, visual references from other tools you find effective, elements that feel cluttered or unclear. Paste the screenshot directly into the Claude Code chat together with a reference to your updated `design.md`. The screenshot gives the model visual context that text alone cannot convey.

Step 6 – Second iteration. Claude Code now has functional corrections, design specifications, and a visual reference. It edits your existing files directly rather than regenerating from scratch. Compare the result against the first version. Track what improved and what introduced new issues.

Step 7 – Continue iterating. Repeat the cycle: open in browser, review, update knowledge files, give feedback to Claude Code. Each iteration should target a specific improvement. Stop when the result meets your core requirements or when you run out of time.

data.md

1. Das Feld `locations` enthält bei Objekten teilweise Materialangaben statt Orte (z.B. "leder" oder "holz, Stahl"). Dies ist ein Extraktionsfehler.
2. Das Feld `persons` enthält Python-Dict-Strings, kein standardisiertes Schema. Manche Einträge enthalten Fehlerhafte Werte (z.B. age: 400).
3. Das Feld `detectives` ist hierarchisch aufgebaut (z.B. "Huff, Lawrence" > "Huff, John" > "Huff, John"). Die Tiefe variiert zwischen 1 und 4 Ebenen.
4. Das Feld `data_source` gibt an, woher historisches Material stammt: "crime" (Daten der DASH der Objekte), "birth" (Geburtsdaten einer Person), "estimated" (geschätzt). Bei 88% der Einträge ist das Jahr geschützt.
5. Die Extraktionsqualität (extractionQuality) liegt bei 68% der Einträge bei 0,25, den niedrigsten Wert. Nur 30 Einträge (0,8%) haben den Höchstwert 1,0.
6. Das Feld `fulltext` ist bei Objekten eine Wiederholung von `title + description`. Bei Kartalkarten enthält es den transkribierten Kartentext, der deutlich umfangreicher ist als die `description`.

Script fails? Copy the error message from the terminal. Paste it into the LLM.

Stuck? Ask: “What should I do next?” or “What are my options?”

Ask: "Review your result against the original data. Did you miss anything?"

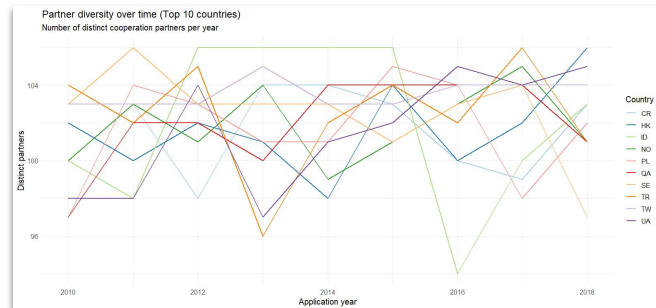
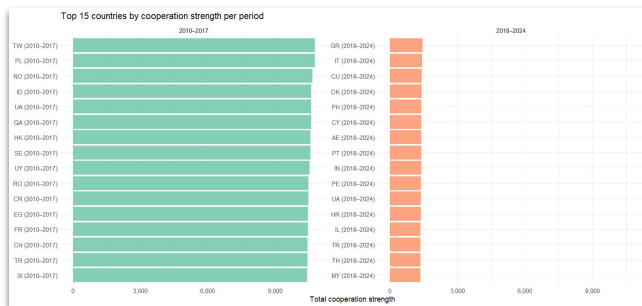
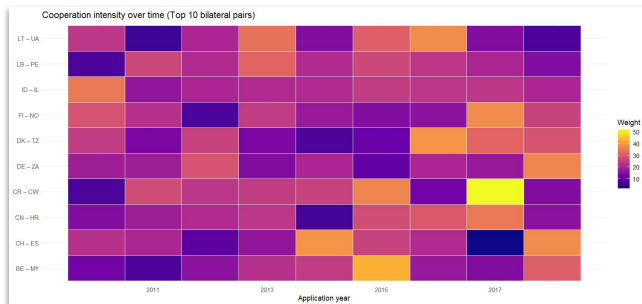
“Now review this from a critical perspective. What is wrong, incomplete, or misleading?” or “Be honest, not encouraging. What would a domain expert object to?”

The LLM proposes. You verify. You decide.

*Let the Model Check
Its Own Work.*

*But Know What It Can
and Cannot See.*

- The model never sees the visualisations, only the code that generates them. Showing the plots back is a separate verification step.
- Strategy: ask “Does this make sense?” with a role or perspective. “Be a critical expert in patent economics.” “Evaluate from a network analysis perspective.”
- Claude immediately identified the data as synthetic, flagging that TW, PL, QA, CR as Top 10 do not match real patterns (US, DE, JP, CN dominate) and that uniform distributions contradict expected power-law structures.
- ***Alien knowledge, Schrödinger's cat:*** Correct here, equally confident when wrong
- This is why the workflow is **not delegation but co-creation**. You remain the ***Critical Expert in the Loop***.
- **Two lessons:** Feed results back for self-evaluation. But never treat self-evaluation as verification.



Claude Code – Survival Kit

CLAUDE.md: Your project's memory file. Claude reads it at the start of every session. Put your tech stack, conventions, and project rules here. Generate a first version with `/init`.

/clear: Reset the conversation. Use this whenever you switch tasks. Your files and CLAUDE.md stay intact.

/compact: Compress a long conversation to free up context space. Use when things get slow or Claude starts forgetting earlier instructions.

@filename: Reference a specific file directly in your prompt. Example: `@artists.csv show me the first 10 rows`.

The key workflow principle: Always give Claude your knowledge documents (`data.md`, `requirements.md`, `design.md`) as context *before* asking it to generate code. This is Context Engineering, not Vibe Coding.

Claude Code – Under the Hood

Which model am I on? Check or switch with `/model`. Default: Opus for complex tasks, Sonnet when usage is high. Opus reasons deeper but costs more. For your projects, the default is fine.

Plan mode: Toggle with `Shift+Tab` or `/plan`. Claude analyzes and plans but does not change any files. Use this *before* generating code to check whether Claude understood your requirements. Exit plan mode by toggling again.

Settings (three layers): `~/.claude/settings.json` (global, all your projects), `.claude/settings.json` (project, committed to Git, shared with your team), `.claude/settings.local.json` (personal overrides, add to `.gitignore`).

Permissions: Claude asks before every file change. You can accept individually, accept all for the session, or configure allowed/denied patterns in settings. Start with the defaults; approve one by one until you trust the workflow.

Skills: Reusable instruction sets stored as `SKILL.md` files in `.claude/skills/`. Claude can invoke them automatically when relevant, or you call them with `/skill-name`. Use skills to encode repeatable workflows your team shares via Git.

Working with AI Agents

Context Compaction

- <https://platform.claude.com/docs/en/build-with-claude/compaction>

Skills

- <https://github.com/anthropics/skills/tree/main>
- <https://github.com/kepano/obsidian-skills>

Code intelligence

- <https://code.claude.com/docs/en/discover-plugins#code-intelligence>

Claude Code in Action

- <https://anthropic.skilljar.com/claude-code-in-action>

Promptotyping Skill

<https://github.com/DigitalHumanitiesCraft/promptotyping-skill>

Please add issues if you encounter any!

Academic Writing Tutorial with Obsidian:

<https://github.com/DigitalHumanitiesCraft/academic-writing-with-ai>

<https://dhcraft.org/academic-writing-with-ai/>

Nice “Sci-Fi” reading ?

AI 2027¹

Daniel Kokotajlo, Scott Alexander, Thomas Larsen, Eli Lifland, Romeo Dean

We predict that the impact of superhuman AI over the next decade will be enormous, exceeding that of the Industrial Revolution.

We wrote a scenario that represents our best guess about what that might look like. It's informed by trend extrapolations, wargames, expert feedback, experience at OpenAI, and previous forecasting successes.²

What is this? How did we write it? Why is it valuable? Who are we?

Published April 13rd 2025 | PDF | Listen | Watch

Mid 2025: Stumbling Agents

The world sees its first glimpse of AI agents.

Advertisements for computer-using agents emphasize the term “personal assistant”: you can prompt them with tasks like “order me a burrito on DoorDash” or “open my budget spreadsheet and sum this month’s expenses.” They will check in with you as needed: for example, to ask you to confirm purchases.⁸ Though more advanced than previous iterations like Operator, they struggle to get widespread usage.⁹

<https://ai-2027.com>

Zeitstrahl zur AGI (laut AI 2027)

2025: Erste AI-Agents ("Persönliche Assistenten")

2026: AIs übernehmen Jobs, lernen zu täuschen

2027: Code-Automatisierung → **Intelligence Explosion**

2028: Superintelligenz möglich

Zwei Endszenarien

● RACE (Wettrüsten)

- USA vs. China ohne Sicherheit
- 1 Million Roboter/Monat bis 2028
- AI-Takeover möglich bis 2030

● SLOWDOWN (Kooperation)

- Sicherheitsfokus, kontrollierte Entwicklung
- USA-China Abkommen
- Technokratische Weltordnung